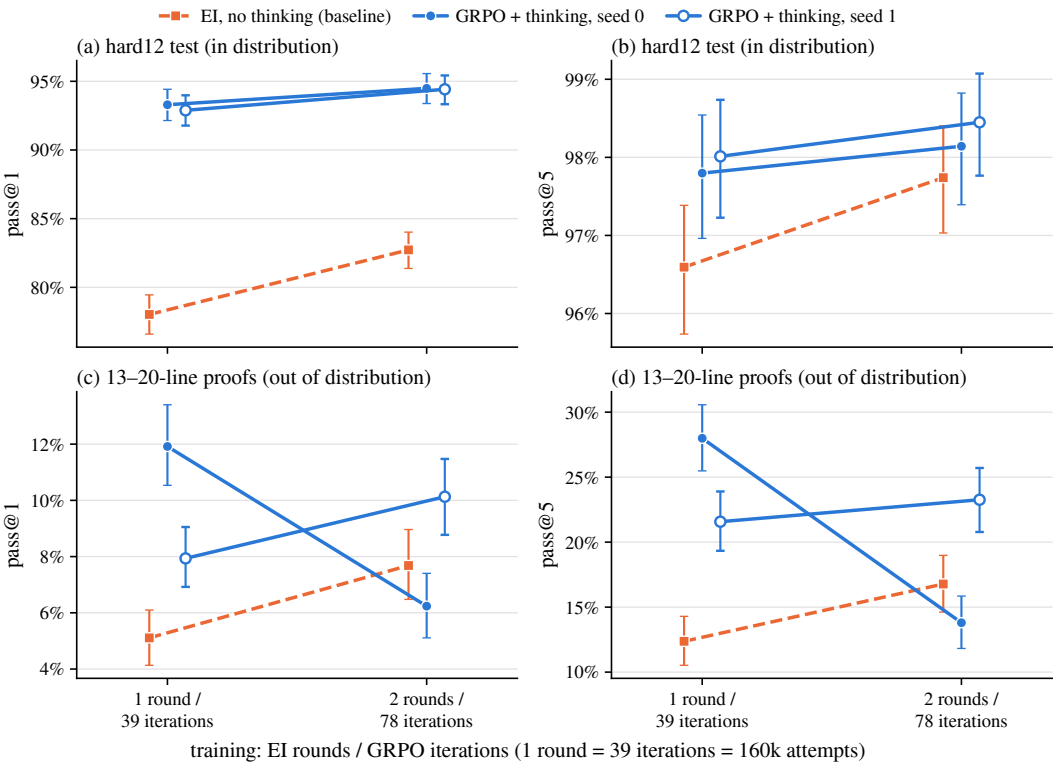
EI, no thinking (baseline)
GRPO + thinking, seed 0
GRPO + thinking, seed 1
(a) hard12 test (in distribution)
(b) hard12 test (in distribution)
(c) 13–20-line proofs (out of distribution)
(d) 13–20-line proofs (out of distribution)
pass@1
pass@5
95%
90%
85%
80%
99%
98%
97%
96%
12%
10%
8%
6%
4%
30%
25%
20%
15%
10%
1 round /
39 iterations
2 rounds /
78 iterations
training: EI rounds / GRPO iterations (1 round = 39 iterations = 160k attempts)